

THREAD: Joint 2D-3D Generation of Egocentric Hand-Object Interactions

Journal:	*SCIENCE CHINA Information Sciences*
Manuscript ID	SCIS-2026-0810
Manuscript Type:	Special Topic: Spatial Intelligence (2026)
Date Submitted by the Author:	31-Mar-2026
Complete List of Authors:	Han, Guangyi; University of Science and Technology of China Zhai, Wei; University of Science and Technology of China Yang, Yuhang; University of Science and Technology of China, Automation Wang, Zining; University of Science and Technology of China Cao, Yang; University of Science and Technology of China Zha, Zheng-Jun; University of Science and Technology of China
Keywords:	Spatial intelligence, Hand-object interaction, Joint 2D-3D generation, Affordance reasoning, Egocentric vision, Diffusion models
Speciality:	Computer Vision\Image Processing < Computer Science & Technology

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THREAD: Joint 2D-3D Generation of Egocentric Hand-Object Interactions

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Abstract Human spatial intelligence is fundamentally rooted in the ability to mentalize future scenes and orchestrate precise physical interactions to realize them. At the heart of this capability lies Hand-Object Interaction (HOI), a complex task that requires a seamless blend of egocentric visual imagination and 3D geometric reasoning. However, existing computational models for HOI face significant challenges due to the inherent disparity between modalities: 2D models often lack geometric consistency, while 3D pipelines struggle to capture global scene context. This creates a cyclic dependency that demands high-quality mutual guidance, further complicated by the efficiency bottlenecks of iterative 3D optimization during joint training. To bridge these gaps, we propose THREAD, a framework leveraging egocentric visual priors for the novel task of Joint 2D-3D Egocentric HOI Generation. Our approach establishes a bidirectional coupling that enhances 2D visual imagination and 3D geometric constraints through a progressive generation pipeline. Specifically, THREAD resolves the cyclic dependency via a prior-guided cold-start mechanism, utilizes predicted interaction heatmaps to constrain the high-dimensional 3D search space, and employs topology-aware semantic points to facilitate joint training. This formulation enables the synchronous optimization of 2D visual and 3D structural diffusion branches. Extensive experiments demonstrate that THREAD achieves an effective alignment between visual realism and physical plausibility, significantly outperforming existing separate 2D and 3D synthesis methods. The project page is here.

Keywords spatial intelligence, hand-object interaction, joint 2D-3D generation, affordance reasoning, egocentric vision, diffusion models

Citation THREAD: Joint 2D-3D Generation of Egocentric Hand-Object Interactions. Sci China Inf Sci, for review

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1 Introduction

The ability to observe a current physical state, visualize a desired future state of an object, and intuitively deduce the required actions to achieve that specific transition is a core aspect of spatial intelligence [1–4]. Rather than merely memorizing continuous motion trajectories, humans predict the discrete evolution of states governed by physical constraints and object affordances, grounding continuous action sequences through state transitions between key observations [5–7]. In the context of embodied artificial intelligence, equipping agents with this ability remains a critical challenge [8]. Recently, generative models trained on large-scale video datasets have shown strong capabilities in modeling rich visual dynamics and temporal patterns from data [9]. Consequently, recent works [10, 11] seeks to combine these powerful 2D priors with 3D representations to synthesize realistic interactions.

Despite this promising direction, current joint 2D-3D HOI paradigms remain limited. In 3D HOI, pure motion generation pipelines are constrained by limited large-scale interaction data [12–14]. Furthermore, these methods struggle in unconstrained environments, as they tend to overfit local contact patterns [15, 16] and lack explicit modeling of global scene context [17, 18]. Meanwhile, efforts to incorporate 2D priors into 3D generation [19] mainly focus on static appearance synthesis rather than modeling dynamic interaction constraints. Methods that extract 3D affordances or interaction regions from 2D inputs [20] remain limited to static representations and lack explicit modeling of scene and interaction dynamics. From the perspective of general 2D generation, image and video diffusion models excel at high-fidelity appearance synthesis [21, 22], but lack explicit modeling of spatial geometry and physical structure. Attempts to introduce spatial awareness in 2D HOI pipelines rely on sparse control signals, such as bounding boxes [23] or 2D hand skeletons [24]. Such sparsity limits the modeling of fine-grained geometry and complex hand-object occlusions [25], hindering precise spatial control for dexterous interactions.

To bridge these gaps, we formalize the task of **Joint 2D-3D Egocentric HOI Generation** towards desired object states. Given an initial observation and the object's current and target spatial configurations, we jointly generate the corresponding 3D interaction and its 2D visual observation. However, this joint formulation faces three core challenges. First, a cyclic dependency arises during the joint generation process because 3D geometric synthesis relies on 2D visual cues, while 2D appearance generation depends on 3D spatial conditions. This interdependency

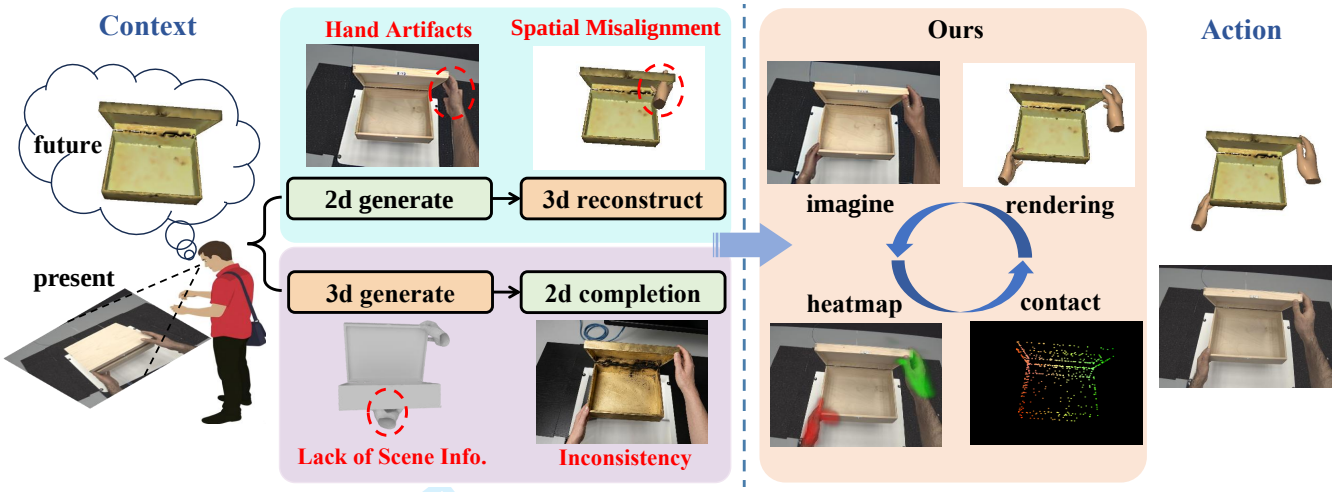


Figure 1 Synthesizing subsequent HOI given initial observations and target states reveals severe bottlenecks in conventional cascaded pipelines. Specifically, performing 2D generation prior to 3D reconstruction frequently produces visual anatomical artifacts and subsequent spatial misalignments. Conversely, initially generating 3D structures before 2D completion lacks macroscopic scene understanding, causing physically implausible spatial configurations (e.g., hands penetrating tabletops) and identity shifts during visual synthesis. To address these limitations, THREAD establishes a unified cycle across intermediate heatmaps, spatial contacts, dense 3D renderings, and visual imagination, mutually enhancing 2D visual priors and 3D geometric constraints.

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presents a cold-start challenge, suggesting that the initialization process can benefit from leveraging large-scale pretrained vision models [26] . Second, the two modalities exhibit complementary deficiencies. Pure 3D generation lacks global scene understanding, often resulting in physically implausible configurations [18]. Conversely, 2D generation suffers from weak spatial control. We observe that extracting explicit affordance constraints from dual 2D priors, namely the spatial semantics encoded by pretrained vision transformers [26,27] and the generative manifold of latent diffusion models [28], provides useful guidance for 3D reasoning, while utilizing dense 3D representations conversely provides the robust structural guidance missing in 2D synthesis. Third, the high computational cost of traditional iterative optimization for 3D anatomical parameters hinders efficient joint training with 2D modalities. Directly optimizing anatomical parameters places the hand and the object in mismatched spaces, whereas optimizing only sparse anchor regions frequently forces unnatural fingertip contacts or severe penetrations. Therefore, we develop an intermediate spatial representation and an efficient mapping strategy to achieve efficient parameter recovery without iterative optimization.

To tackle the proposed task and implement these insights, we propose THREAD, a unified framework establishing a bidirectional coupling to enhance 2D visual imagination and 3D geometric constraints. To resolve the cyclic dependency, THREAD introduces a prior-guided cold-start mechanism that extracts rich visual features from the initial observation using a pretrained DINOv2 model [26]. Utilizing these features, the model predicts a fine-grained 2D contact heatmap, which effectively imposes explicit affordance constraints to regularize the high-dimensional 3D search space. Concurrently, the synthesized dense 3D renderings and depth maps serve as explicit spatial conditions for the 2D diffusion branch, overcoming the weak controls of visual generation [29]. To address the spatial mismatch during joint optimization, the 3D diffusion process operates directly on topology-aware semantic points, aligning the hand with the object geometry in a unified space through optimization driven by a Score Distillation Sampling (SDS) loss [30]. Subsequently, we introduce a non-iterative inverse kinematics algorithm to recover the anatomical hand structure. This specialized design preserves the robust physical constraints of structured representations while eliminating the computational burden of slow iterative optimizations.

In summary, the main contributions of this paper are threefold. (1). We formulate a novel paradigm for HOI synthesis that models discrete state transitions, establishing a bidirectionally coupled architecture to mutually enhance the 2D and 3D generation outcomes. (2). We propose a specialized semantic point representation for the interacting hand accompanied by a novel non-iterative inverse kinematics algorithm, which prevents representation mismatches while significantly accelerating structured parameter recovery. (3). By integrating Score Distillation Sampling and explicit affordance constraints into a unified generation process, our method effectively promotes physical consistency and macroscopic scene understanding, significantly outperforming current methods in generating highly realistic and spatially accurate interactions.

2 Related Work

2.1 3D Hand-Object Interaction Generation

Synthesizing realistic 3D interactions between human hands and objects represents a fundamental challenge in computer vision and embodied artificial intelligence [31]. Recent advancements have transitioned from stable grasp estimation [32,33] to the generation of dynamic spatial manipulations [34,35]. To formulate these complex interactions, diverse 3D representations exist to model the dexterous hand. These representations span from MANO parametric spaces [36], axial angle and joint coordinates to continuous occupancy fields and point clouds [32,33,37,38]. However, directly optimizing structural parameters often places the hand and the object in mismatched spatial domains [39]. Furthermore, relying on sparse anchor regions to alleviate this mismatch frequently forces unnatural kinematic behaviors and severe interpenetration [40,41]. Beyond the challenges of local geometric alignment, these pure 3D methods fundamentally struggle to capture broader scene semantics. The generation pipelines typically degenerate into the local optimization of hand-object contact topologies [17,18]. Lacking a macroscopic understanding of non-template object affordances and environmental contexts [42], these models lose their applicability in wild scenarios. This inherent limitation highlights the necessity of incorporating 2D external knowledge to guide the interaction generation.

2.2 From 2D Priors to 3D Interaction Modeling

Bridging the gap between visual spaces and physical environments represents a rapidly evolving research area. The emergence of large-scale generative models has significantly advanced visual synthesis, demonstrating a strong ability to generate high-fidelity visuals [43]. For instance, foundational vision models trained on large-scale image datasets have acquired rich semantic and geometric priors [26, 44] . These learned priors enable the capture of spatial transformations and the deduction of plausible interaction regions [45]. Consequently, these features support downstream tasks that include pose estimation [46, 47] and interaction hotspot detection. Building on this foundation, several studies extract 3D affordances and interaction regions directly from 2D images or videos [48,49]. However, these affordances remain primarily limited to static representations [48]. They lack explicit modeling of dynamic scene contexts and interaction mechanisms [50]. In the context of physical dynamics, video generation networks can capture temporal patterns of object behaviors [51]. These models learn motion dynamics from sequential observations [52]. Despite this progress, achieving precise spatial manipulation remains challenging within pure 2D visual spaces.

2.3 Bidirectional 2D-3D Interaction Modeling

Attempts to inject spatial awareness into pure 2D generation pipelines typically rely on sparse 3D conditions, such as bounding boxes or discrete points [11, 23]. Particularly for the guidance of interacting hands, current methods predominantly utilize sparse skeletons rather than dense 3D representations [24, 53, 54]. This structural sparsity limits fine-grained guidance for interaction regions and complicates the resolution of intricate geometric details and occlusion relationships [55]. Beyond sparse conditioning, recent approaches incorporate richer geometric signals, such as depth or normal maps, to enhance spatial consistency in 2D synthesis [25, 56, 57]. Conversely, prevalent 3D generation frameworks extract semantic priors from 2D models via score distillation sampling [58–60]. These approaches primarily focus on synthesizing high-quality static appearances and often lack explicit modeling of dynamic interactions [61]. More recent paradigms further explore direct mappings from 2D features to structured 3D representations, enabling reconstruction from image observations [19, 62].

Despite progress in both directions, modeling hand-object interaction requires jointly reasoning about visual fidelity and physical constraints [63,64]. 2D visual cues provide texture and contact evidence, while 3D representations impose kinematic constraints. However, effectively integrating these complementary factors remains challenging for achieving physically plausible interactions [24, 65].

3 Method

In this section, we present THREAD, a framework for the joint 2D-3D generation of first-person hand-object interactions. This work aims to generate realistic first-person hand-object interactions by predicting the imagined future state $\hat{I}_1$ and the potential hand action $\hat{P}_1^H$, given all information about the current state of the agent (I_0, $\hat{P}_0^H$, $\hat{P}_0^{ego}$) and the target poses of both the object and the agent (P_1^{obj}, $\hat{P}_1^{ego}$). To tackle this problem, we propose a three-module framework, as illustrated in Figure 2. Specifically, the Prior-Guided 2D State Synthesis and

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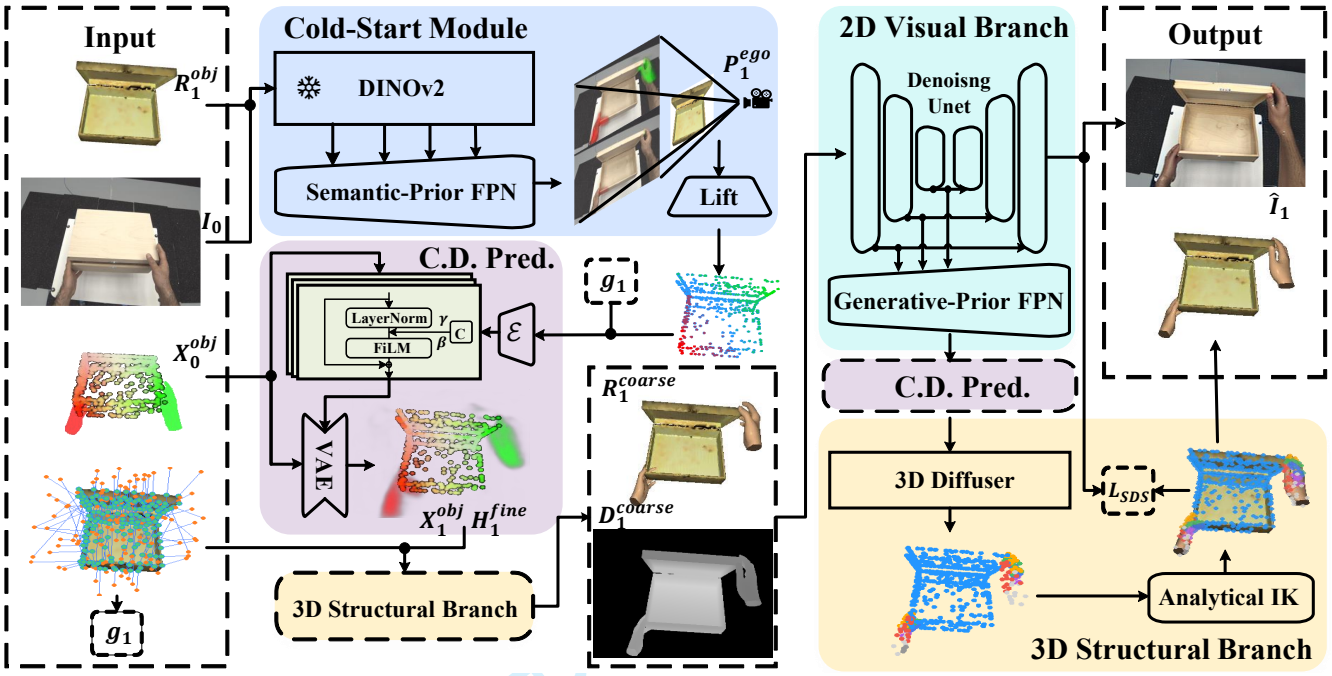


Figure 2 Overview of the proposed framework for the joint 2D-3D generation of first-person hand-object interactions. The semantic-prior cold-start module predicts an interaction heatmap from the initial observation to resolve the cyclic dependency between appearance synthesis and pose estimation. This prediction is lifted into a 3D contact representation to formulate explicit structural conditions. The 2D generative-prior branch then synthesizes the target image and extracts a spatial interaction heatmap from the diffusion decoder. Guided by these visual and geometric constraints, the 3D branch predicts point-cloud-based hand motions, which are converted into parametric mesh representations via an efficient inverse kinematics algorithm. Blue points are associated with the object, red with the left hand, and green with the right hand. "C.D. Pred." means "Contact Distance Prediction", and "IK" means "Inverse Kinematics".

Hotspot Extraction module (Section 3.1) synthesizes high-fidelity target frames and extracts interaction hotspots by leveraging dual 2D visual priors, namely the spatial semantics and the generative manifold. The Heatmap-to-Contact Lifting module (Section 3.2) lifts the 2D interaction heatmap to a 3D contact representation in the Basis Point Set (BPS) space [66], providing explicit geometric constraints for 3D hand pose synthesis. Finally, the Efficient Point-Cloud-Based Hand Motion Generation module (Section 3.3) synthesizes accurate 3D hand poses via a diffusion-based generative model, followed by an efficient inverse kinematics algorithm to recover the parametric MANO representation.

3.1 Prior-Guided 2D State Synthesis and Hotspot Extraction

To extract explicit hotspot constraints and generate the target visual appearance from the dual 2D priors, we design a framework that independently leverages the spatial semantics encoded by a pre-trained vision transformer and the generative manifold of a latent diffusion model [28]. Specifically, both pathways incorporate a Feature Pyramid Network (FPN) [67] architecture to predict spatial interaction heatmaps. The semantic prior effectively resolves the initial geometric ambiguity to break the cyclic dependency, while the generative prior synthesizes the final image and extracts contact regions.

Semantic Prior for Cold-Start Initialization. To address the cyclic dependency and the resulting cold-start problem, our framework must initialize the interaction states before establishing the full 2D-3D loop. We circumvent this dilemma by leveraging the pre-trained vision transformer to extract an initial spatial interaction distribution directly from 2D visual semantics.

We utilize a frozen DINOv2 encoder [26] to extract multi-scale visual features from the initial observation I_0 and the object-only rendering $R_1^{obj} \in \mathbb{R}^{3\times H\times W}$. DINOv2, trained on massive image datasets through self-supervised learning, has learned rich semantic and geometric priors that can capture spatial transformations and infer plausible interaction regions. Features extracted from selected transformer layers, denoted as $\{f_0^{(l)}, f_1^{(l)}\}_{l\in\mathcal{L}}$, provide a comprehensive representation spanning semantic context to granular spatial details [27]. To capture cross-frame interaction cues, the dual-branch features at each layer are fused via a correlation-aware aggregation. This operation concatenates the original features, their absolute differences, and their Hadamard products, explicitly encoding

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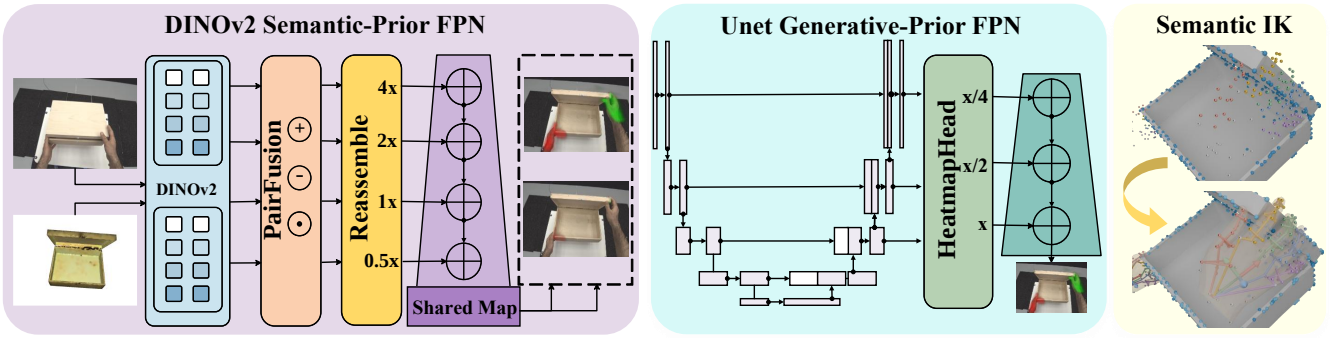


Figure 3 Detailed architecture of the dual-prior hotspot extraction and semantic inverse kinematics modules. Left: the semantic-prior pathway leverages a pretrained vision model to predict an interaction heatmap and an occlusion map for cold-start initialization. Middle: the generative-prior pathway extracts a spatial interaction heatmap from the diffusion decoder via an FPN head. Right: the semantic anchor-based analytical inverse kinematics pipeline recovers MANO parameters from predicted point clouds, where blue points are associated with the object, red with the left hand, and green with the right hand.

structural transformations. A dedicated FPN decodes these aggregated representations through top-down pathways and lateral connections. The network ultimately outputs an interaction heatmap $H_1^{dino} \in [0,1]^{H' \times W'}$, which estimates potential manipulation hotspots based on the intrinsic affordance of the object.

Generative Prior for High-Fidelity Synthesis. Building upon the cold-start estimation, the target appearance is synthesized using a latent diffusion formulation. The current observation I_0, the coarse interaction rendering $R_1^{coarse} \in \mathbb{R}^{3\times H\times W}$, and the associated depth map $D_1^{coarse} \in \mathbb{R}^{1\times H\times W}$ are embedded into a latent space via a shared VAE encoder $\mathcal{E}$ [68]. The denoising condition $c \in \mathbb{R}^{(3C)\times h\times w}$ is constructed by concatenating the respective latent representations along the channel dimension.

To further guide the generation towards the desired interaction intent, following [69], we utilize a frozen CLIP [44] visual encoder $\tau : \mathbb{R}^{3\times H\times W} \to \mathbb{R}^{D_{clip}}$ to extract global semantic embeddings from R_1^{coarse}, where D_{clip} denotes the CLIP embedding dimension. These semantic tokens are injected into the UNet via cross-attention layers at multiple resolution levels, facilitating a deep alignment between visual features and interaction intent. At each denoising step, the noisy latent z_t is processed through the encoder path, where features are progressively downsampled and enriched with semantic information via cross-attention with $\tau(R_1^{coarse})$. The decoder path then upsamples these features while incorporating skip connections from the encoder, ultimately predicting the noise ϵ added to the clean latent. The denoising objective at timestep t is formulated as:

$$\mathcal{L}_{2D} = \mathbb{E}_{z,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(z_t \oplus c; t, \tau(R_1^{coarse}))\|^2\right], \tag{1}$$

where $\oplus$ denotes channel-wise concatenation, and $t \sim \mathcal{U}(1,T)$ is uniformly sampled from the diffusion timesteps.

To explicitly extract interaction hotspots from the generated appearance, we attach a Feature Pyramid Network (FPN) head to the UNet decoder. The FPN aggregates multi-scale features $\{f_i\}_{i=1}^4$ from different resolution levels of the UNet decoder, where $f_i \in \mathbb{R}^{C_i\times h_i\times w_i}$ with progressively increasing spatial resolutions. These features are first upsampled to a common resolution and fused via element-wise addition, then passed through a lightweight convolutional head to produce the interaction heatmap $H_1^{diff} \in [0,1]^{H'\times W'}$, where $H'\times W'$ denotes the heatmap resolution. The heatmap prediction is supervised by a weighted binary cross-entropy loss: $\mathcal{L}_{hm} = \text{BCE}_w(H_1^{diff}, H^{gt})$, where $H^{gt} \in \{0,1\}^{H'\times W'}$ is the ground truth heatmap, and $\alpha > 1$ is a positive weight to emphasize contact regions.

3.2 Heatmap-to-Contact Lifting

This module lifts the 2D interaction probabilities into a 3D geometric representation of hand-object distances. By parameterizing these distances within a Basis Point Set (BPS) space, it establishes explicit spatial constraints for the subsequent 3D hand pose synthesis.

Geometric Parameterization via Basis Point Sets. To effectively represent the contact distribution on rigid objects, along with articulated objects, we adopt the Basis Point Set (BPS) encoding [66]. BPS represents an object by measuring the signed distances from a set of fixed basis points $\{\mathbf{b}_i\}_{i=1}^{N_{bps}}$ to the object surface. For an articulated object with K parts, each part p is associated with a local BPS centered at the part's coordinate frame [70]. Given

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the object point cloud $\mathcal{V}_p = \{\mathbf{v}_j\}_{j=1}^{N_p}$ for part p, the BPS encoding is computed as:

$$\mathbf{x}_i^p = \frac{1}{s_o} \arg\min_{\mathbf{v} \in \mathcal{V}_p} \left\| \frac{\mathbf{v}}{s_o} - \mathbf{b}_i \right\| - \mathbf{b}_i, \tag{2}$$

where s_o is the object scale factor for normalization, and $\mathbf{x}_i^p \in \mathbb{R}^3$ represents the displacement vector from basis point $\mathbf{b}_i$ to its nearest surface point on part p. The complete BPS encoding for the object is the concatenation $\mathbf{X} = [\mathbf{x}_1^1, \ldots, \mathbf{x}_{N_{bps}}^K] \in \mathbb{R}^{3 \times K \cdot N_{bps}}$. The BPS representation offers several advantages for hand-object interaction modeling: it provides a compact and structured encoding that naturally handles articulated objects, the basis points are strategically distributed to emphasize object edges where contacts are most likely to occur, and the representation can be efficiently computed via nearest-neighbor queries.

Residual Contact Distance Prediction. To lift the 2D heatmap H_1^{dino} into the 3D BPS space, we project each basis point $\mathbf{b}_i$ onto the image plane using the known camera intrinsics and the target object pose. The corresponding contact probability $h_i \in [0, 1]$ is then extracted via bilinear interpolation at the projected pixel coordinates. This operation yields a per-basis-point probability distribution $\{h_i\}_{i=1}^{K \cdot N_{bps}}$, indicating the likelihood of hand-object contact across the structural surface.

To accurately infer the target interaction state, we employ a Feature-wise Linear Modulation (FiLM) [71] residual architecture to predict the contact distance change $\Delta\mathbf{D} \in \mathbb{R}^{K \cdot N_{bps}}$ from the current state to the target state. The predictor takes two streams as input: a base stream encoding the anchor frame's contact distance $\mathbf{D}_0 \in \mathbb{R}^{K \cdot N_{bps}}$, and a condition stream encoding the 3D BPS coordinates concatenated with the lifted heatmap probabilities $[\mathbf{X}; \{h_i\}] \in \mathbb{R}^{(3+C_{cond}) \times K \cdot N_{bps}}$, where $C_{cond} = 4$ represents the per-hand occlusion probabilities. The FiLM mechanism modulates the base features via learned affine transformations conditioned on the heatmap:

$$\mathbf{h}_{base} \leftarrow \mathbf{h}_{base} + f_{FiLM}(\mathbf{h}_{base} \odot (1 + \gamma) + \beta), \tag{3}$$

where $\gamma, \beta = \text{MLP}_{cond}([\mathbf{X}; \{h_i\}])$ are the modulation parameters. The final prediction is formulated as a residual: $\mathbf{D}_1 = \mathbf{D}_0 + \Delta\mathbf{D}$, supervised by SmoothL1 loss. This residual formulation enables the model to focus on the contact state transitions rather than absolute distances, improving generalization across diverse interaction patterns.

3.3 Efficient Point-Cloud-Based Hand Motion Generation

This module synthesizes accurate 3D hand poses in the point cloud space via a diffusion-based generative model, followed by an efficient inverse kinematics algorithm to recover the parametric MANO representation, achieving a deep alignment of interaction intention and structural integrity.

Semantic Anchor-Based Analytical Inverse Kinematics. To bridge the gap between the predicted action-space point cloud and the parametric MANO model [36], we propose an efficient analytical inverse kinematics pipeline. The MANO model parameterizes a hand mesh using a low-dimensional vector $\Theta = (\boldsymbol{\phi}, \boldsymbol{\theta}, \mathbf{t}, \boldsymbol{\beta})$, which represents the global orientation, joint rotations, root translation, and shape coefficients, respectively. Given the predicted point cloud $\hat{\mathcal{P}}$ containing N_H points per hand, we recover Θ through structured semantic anchor selection on the canonical MANO mesh $\mathcal{M}_0$. Instead of employing heuristic sampling, we partition the N_H points into four semantically distinct groups based on anatomical significance to strictly preserve kinematic constraints. First, we define 45 segment ring anchors by selecting three non-collinear vertices $\{v_s^{mid}, v_s^+, v_s^-\}$ on the cross-sectional ring perpendicular to the bone axis for each of the 15 phalanx segments $s \in \mathcal{S}$. This structural configuration enables robust local frame estimation via the standard Kabsch algorithm [72], which minimizes the distance between the rotated rest-pose coordinates $\bar{\mathbf{v}}$ and the observed coordinates $\hat{\mathbf{v}}$:

$$\mathbf{F}_s = \arg\min_{\mathbf{R} \in \text{SO}(3)} \sum_{j \in \{mid, +, -\}} \|\mathbf{R}\bar{\mathbf{v}}_s^j - \hat{\mathbf{v}}_s^j\|^2. \tag{4}$$

Second, we specify 40 joint and tip pair anchors by selecting two diametrically opposed vertices $\{v^+, v^-\}$ at each of the 20 articulation sites, providing a robust joint center estimate via simple averaging: $\mathbf{c}_k = \frac{1}{2}(\hat{\mathbf{v}}_k^+ + \hat{\mathbf{v}}_k^-)$. Finally, the remaining points are allocated to 6 wrist cuff anchors surrounding the wrist circumference to anchor the global root frame, and 9 palm surface anchors selected via farthest point sampling to enhance overall spatial coverage.

Direct Single-Pass Parameter Recovery. Building upon these semantic anchors, we introduce a direct, single-pass estimation process to recover the pose parameters sequentially without iterative optimization. Assuming

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a known shape $\boldsymbol{\beta}$, we first estimate an isotropic scale factor α from the median ratio of the rest-pose bone lengths to the observed bone lengths:

$$\alpha = \text{median}_{s\in\mathcal{S}} \frac{\|\bar{\mathbf{c}}_s^{dist} - \bar{\mathbf{c}}_s^{prox}\|}{\|\hat{\mathbf{c}}_s^{dist} - \hat{\mathbf{c}}_s^{prox}\|}. \tag{5}$$

The observed point cloud is then proportionally rescaled to match the canonical dimensions. Next, the global orientation $\mathbf{R}_{root} \in \text{SO}(3)$ is obtained by constructing an orthogonal coordinate frame from five anatomical landmarks: the wrist center and the base positions of the middle, index, and pinky fingers. The resulting rotation matrix $\mathbf{R}_{root} = [\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}]\bar{\mathbf{F}}_{root}^{\top}$ aligns the predicted frame with the rest-pose root frame $\bar{\mathbf{F}}_{root}$. For local articulations $\boldsymbol{\theta}$, the bone axis $\hat{\mathbf{a}}_s$ of each segment is efficiently estimated via weighted Principal Component Analysis on the corresponding finger joint-center chain. The local joint rotations $\mathbf{R}_s^{loc}$ are subsequently calculated by traversing the kinematic tree and computing the minimum-angle rotation:

$$\mathbf{a}_s^{local} = \mathbf{R}_{pa(s)}^{\top}\hat{\mathbf{a}}_s, \quad \mathbf{R}_s^{loc} = \text{RotBetween}(\bar{\mathbf{a}}_s, \mathbf{a}_s^{local}), \tag{6}$$

where $pa(s)$ denotes the parent segment. Finally, the root translation $\mathbf{t}$ is derived through full-mesh consensus by averaging the spatial residuals between the predicted points and the corresponding MANO vertices:

$$\mathbf{t} = \frac{1}{N_H}\sum_{i=1}^{N_H}(\hat{\mathbf{p}}_i - \mathcal{M}(\boldsymbol{\phi}, \boldsymbol{\theta}, \mathbf{0}, \boldsymbol{\beta})_{v_i}). \tag{7}$$

Transformer-Based Diffusion Denoiser. We parameterize the hand pose in an object-centric coordinate frame to disentangle local manipulation from global trajectories. The joint action state $y \in \mathbb{R}^{D_{action}}$ is defined as $y = [y^H; \mathbf{v}^{dir}]$, where $y^H = \{\mathbf{p}_i\}_{i=1}^{N_H}$ represents a sparse set of N_H surface keypoints sampled from the MANO mesh, and $\mathbf{v}^{dir} = \{\mathbf{v}_i\}_{i=1}^{N_H}$ denotes direction vectors originating from the keypoints to their nearest object vertices. These direction vectors explicitly encode both the magnitude and spatial orientation of the hand-object relationship, providing critical geometric cues for contact reasoning. Consequently, the total action dimension is formulated as $D_{action} = 6N_H$. To ensure robust generalization across varying object trajectories, all hand keypoints are represented in the canonical coordinate frame of the object, aligning the articulation axis with the negative z-axis. The Transformer-based denoiser f_ϕ models the hand interaction conditioned on the spatial heatmap $H_1^{spatial}$, the BPS spatial features $\mathbf{X}_1^{obj}$, and the global object state $\mathbf{g}_1$. The noisy action state y_t is linearly projected into N_H spatial tokens, while the diffusion timestep t and the three conditional signals are encoded into 4 auxiliary tokens via independent multi-layer perceptrons. These components are mapped into a unified latent dimension D_{emb} and concatenated to form a sequence $\mathbf{S} \in \mathbb{R}^{(N_H+4)\times D_{emb}}$. A Transformer architecture, comprising L layers of multi-head self-attention and feed-forward networks, processes this sequence to directly predict the uncorrupted action state $\hat{y}_0$ from the first N_H output tokens. The overall network is optimized using a standard diffusion objective:

$$\mathcal{L}_{action} = \mathbb{E}_{y_0,\epsilon,t}\left[\|y_0 - f_\phi(y_t; t, H_1^{spatial}, \mathbf{X}_1^{obj}, \mathbf{g}_1)\|^2\right]. \tag{8}$$

3.4 Bidirectional 2D-3D Coupling and Joint Optimization

Bidirectional 2D-3D Coupling. To enforce mutual consistency between the 2D and 3D modalities, we employ a bidirectional feedback loop where a parameter-shared 3D diffusion model undergoes two forward passes per training iteration. During the cold-start phase, the intermediate 3D hand-object interaction is rendered into 2D RGB and depth maps. These renderings serve as input conditions for the 2D branch, which enables the 2D model to better adapt to the 3D outputs and promotes synergistic optimization. Importantly, the computational graph is explicitly detached in this specific path to prevent gradient flow back to the 3D model. Conversely, in the main joint training branch, the Score Distillation Sampling (SDS) loss provides direct gradient updates to the 3D parameters by treating the 2D diffusion model as a learned prior. Given the predicted 3D hand pose $\hat{y}_0$, we render it alongside the object to obtain a synthetic image $\hat{I}_1 = \text{Render}(\hat{y}_0, \mathbf{T}_1^{obj})$. The SDS loss is computed as [30]:

$$\mathcal{L}_{SDS} = \mathbb{E}_{t,\epsilon}\left[w(t)\|\epsilon - \epsilon_\theta(\mathcal{E}(\hat{I}_1); t, c)\|^2\right], \tag{9}$$

where $w(t)$ is a weighting function that emphasizes certain timesteps, and c denotes the conditioning information for the 2D diffusion model. This loss encourages the 3D pose to generate 2D renderings that are consistent with the learned 2D distribution.

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Progressive Training Strategy. Our framework employs a progressive training protocol, where each module is initially optimized independently before undergoing joint end-to-end refinement. During the initialization phase, the cold-start module is trained to predict interaction heatmaps and contact distance fields using the objective $\mathcal{L}_{cold} = \mathcal{L}_{hm}^{dino} + \lambda_{dist}\mathcal{L}_{dist}$. The pre-trained visual backbone remains frozen to retain its robust feature extraction capabilities. Concurrently, the 2D diffusion branch is trained to synthesize high-fidelity target frames via $\mathcal{L}_{2D} = \mathcal{L}_{2D-diff} + \lambda_{hm}\mathcal{L}_{hm}$, where $\mathcal{L}_{hm}$ evaluates the binary cross-entropy of the predicted heatmaps. Once the individual components converge, we integrate them into a unified optimization framework:

$$\mathcal{L}_{joint} = \lambda_{hm}\mathcal{L}_{hm} + \lambda_{action}\mathcal{L}_{action} + \lambda_{SDS}\mathcal{L}_{SDS} + \lambda_{hm}\mathcal{L}_{hm} + \lambda_{reg}(\mathcal{L}_{contact} + \mathcal{L}_{penetration}). \tag{10}$$

Here, $\mathcal{L}_{action}$ penalizes the 3D point cloud generation errors, while the physical regularization terms enforce plausible contact alignment and prevent hand-object interpenetration. Finally, to ensure superior visual quality for downstream applications, the parameters obtained from the direct single-pass estimation can be optionally refined through a gradient-based post-processing step. This refinement minimizes an auxiliary objective,

$$\mathcal{L}_{refine} = \mathcal{L}_{point} + \lambda_{anat}\mathcal{L}_{anat} + \lambda_{cont}\mathcal{L}_{contact} + \lambda_{pene}\mathcal{L}_{penetration}, \tag{11}$$

which explicitly penalizes anatomically implausible joint articulations and refines fine-grained physical contacts.

4 Experiments

4.1 Experimental Settings

Implementation Details. The framework is trained on 8 NVIDIA 4090 GPUs utilizing mixed-precision optimization. Input images are resized to a resolution of $H \times W = 256 \times 256$ to extract latent representations at a spatial resolution of $h \times w = 32 \times 32$ with $C = 4$ channels. Built upon a pretrained latent diffusion backbone [28], the 2D branch features a UNet architecture with four resolution levels and channel dimensions of $[320, 640, 1280, 1280]$. The 3D pose diffusion model adopts a transformer architecture comprising $L = 8$ layers with an embedding dimension of $D_{emb} = 512$ and 4 attention heads. Interaction geometry is captured by sampling $N_H = 100$ surface points per hand and representing the object via $N_{bps} = 1024$ Basis Point Set (BPS) points. The diffusion process operates over $T = 1000$ training timesteps utilizing a scaled linear DDIM noise schedule [73]. Unless otherwise specified, all models are optimized using the AdamW optimizer.

The 2D diffusion branch is trained for 500 epochs with a learning rate of 10^{-4} while keeping the VAE frozen. Guided by geometric contact cues, the 3D pose diffuser is then optimized within the action space for 200 epochs using a learning rate of 10^{-4}. Initialized with pretrained weights from individual modules, the full framework undergoes joint refinement for 200 epochs at a learning rate of 10^{-5}, wherein the 3D pose diffuser and 2D heatmap modules are updated while the primary 2D diffusion backbone remains frozen to preserve generative priors.

Datasets. The proposed approach is evaluated on the ARCTIC dataset [74], which comprises over 2.1 million frames of egocentric bimanual interactions across 10 object categories, utilizing a standard 80%-20% train-test split. To construct training sequences, a pair generation strategy driven by hand contact state transitions merges anomalous segments shorter than three frames. Frame pairs around these transitions are sampled using a temporal stride of $S = 8$ and a minimum gap of $G_{\min} = 32$, whereas segments shorter than $2G_{\min}$ are split at the midpoint. This configuration yields $18,710$ interaction pairs.

To validate in-the-wild generalization, some evaluations are conducted on the WildO2 dataset [75], which provides unconstrained hoi frame-pairs that introduce challenges in background complexity and interaction variety.

Evaluation Metrics. Evaluating the 2D generation quality involves diverse metrics spanning pixel-level fidelity to high-level semantic realism. Fundamental reconstruction accuracy is measured via Mean Squared Error (MSE), Mean Absolute Error (MAE), Peak Signal-to-Noise Ratio (PSNR), and the Structural Similarity Index Measure (SSIM). Beyond pixel differences, Learned Perceptual Image Patch Similarity (LPIPS) [78] captures human-aligned perceptual fidelity. Furthermore, Fréchet Inception Distance (FID), its spatial variant (sFID), and Inception Score (IS) [79] assess distribution realism and diversity, while CLIPScore [80] quantifies the semantic alignment between visual contents and condition prompts.

For 3D HOI synthesis, the evaluation focuses on geometric precision and physical plausibility. Mean Per-Vertex Position Error (MPVPE) and Mean Per-Joint Position Error (MPJPE) quantify vertex and joint estimation accuracies, whereas Chamfer Distance (CD) evaluates the shape alignment between predicted and ground-truth geometries.

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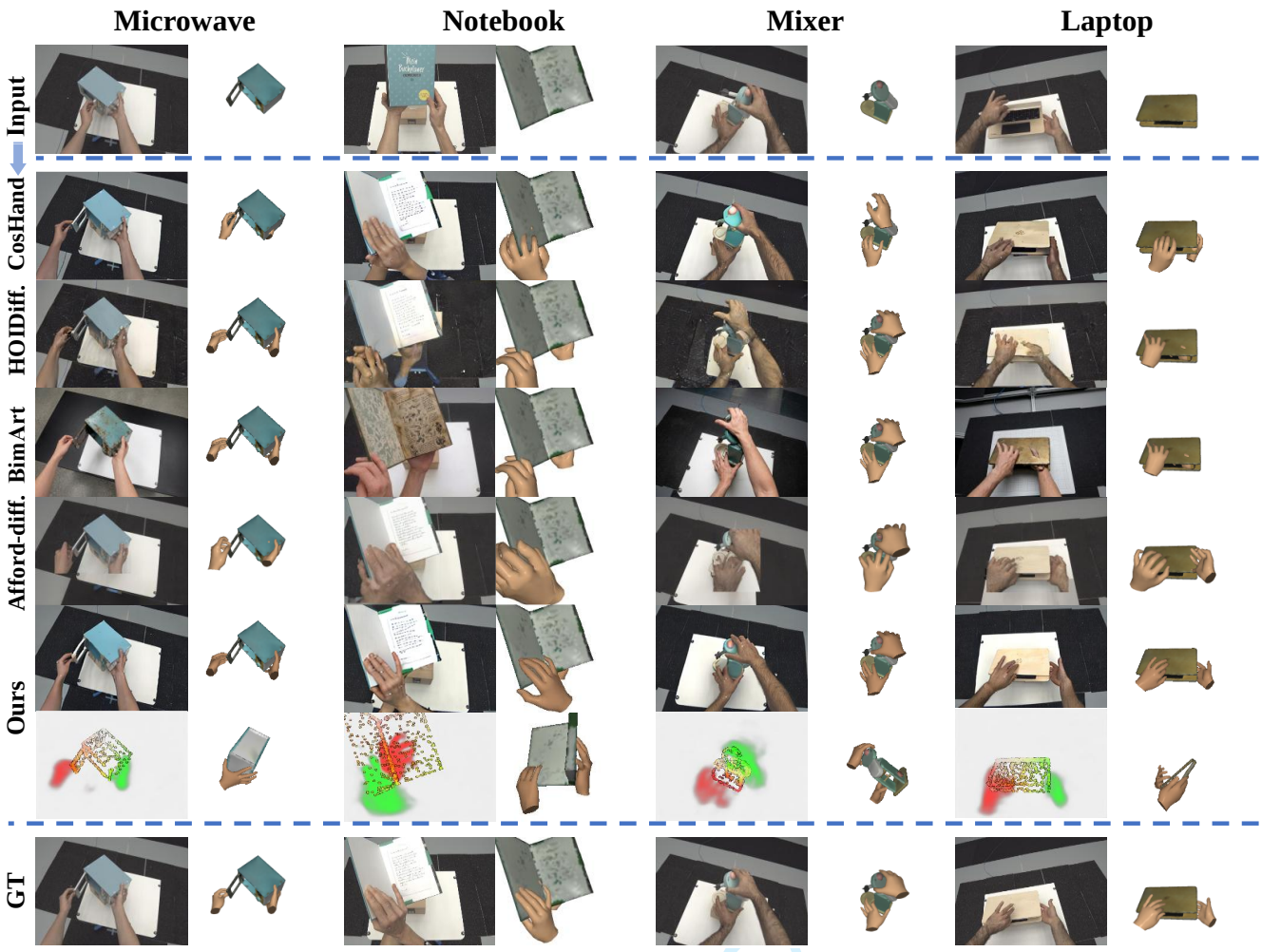


Figure 4 Qualitative results of THREAD on the ARCTIC dataset. Comparisons include: (1) HOIDiff. [76], which utilizes BimArt [70] as the generative frontend; (2) BimArt [70] baseline, employing the general-purpose Nano Banana 2 [77] backend; (3) Afford-diff, where results are re-projected back to the original image space from cropped regions; and (4) CosHand, which takes heatmaps from the cold-start stage as conditional masks.

Physical realism is measured through the Contact Ratio to indicate valid interactions, alongside Penetration Depth (Pen.Depth) and Penetration Volume (Pen.Vol.) to penalize unnatural geometric intersections. To assess macroscopic generative quality, Point-FID (pFID) [81] calculates the Fréchet Inception Distance between generated and target point clouds by extracting spatial features via a pre-trained Point3D feature extractor.

4.2 Comparisons

The selected baseline methods primarily fall into two paradigms, which comprise the 2D-to-3D generation pipeline and the reverse 3D-to-2D pipeline. Representative methods of the first paradigm include CosHand [69], which operates as a diffusion model guided by target-state masks with the source image as conditional input, and Afford-diff [82], which relies on the target background for generation. The adapted Afford-diff extends the original single-hand paradigm into a bimanual framework, utilizes lollipop masks as dual-hand priors, and executes the generation process within a focused crop space. For the second paradigm, BimArt [70] serves as the primary baseline due to the scarcity of approaches addressing articulated object interactions. This framework supports diverse downstream 2D completion schemes, accommodating general 2D generation models like Nano Banana 2 [77] or native 3D-to-2D hand-object completion approaches like HOIDiffusion [76]. To ensure structural rationality, native 2D generation models leverage explicit target-state geometric constraints, which include semantic masks, surface normals, and hand joints.

Since multiple baselines require target background information, a cross-frame state transition pipeline provides consistent background priors through decoupled background restoration, viewpoint transformation, and foreground

Method	PSNR↑	SSIM↑	LPIPS↓	MSE↓	MAE↓	FID↓	sFID↓
HOIDiff.*	$12.836_{\pm 2.727}$	$0.515_{\pm 0.083}$	$0.471_{\pm 0.116}$	$0.062_{\pm 0.034}$	$0.150_{\pm 0.054}$	68.790	0.319
HOIDiff. [76]	$13.552_{\pm 3.095}$	$0.535_{\pm 0.089}$	$0.431_{\pm 0.126}$	$0.055_{\pm 0.034}$	$0.138_{\pm 0.057}$	53.100	0.276
BimArt [70]	$13.018_{\pm 2.241}$	$0.489_{\pm 0.057}$	$0.523_{\pm 0.091}$	$0.057_{\pm 0.029}$	$0.167_{\pm 0.045}$	378.796	1.169
CosHand [69]	$15.680_{\pm 3.521}$	$0.565_{\pm 0.083}$	$0.270_{\pm 0.138}$	$0.037_{\pm 0.033}$	$0.096_{\pm 0.056}$	30.021	0.126
Ours.	$16.187_{\pm 3.533}$	$0.580_{\pm 0.084}$	$0.252_{\pm 0.127}$	$0.033_{\pm 0.029}$	$0.091_{\pm 0.049}$	28.183	0.103

Table 1 Quantitative comparison of 2D generation quality (ARCTIC). Best results are highlighted in gray. HOIDiff.* signifies the variant integrated with our cold-start 3D frontend; HOIDiff and BimArt follow the settings in Figure 4. Afford-diff is excluded as its cropped-space generation leads to domain misalignment.

Method	MPVPE↓	MPJPE↓	CD↓	Contact Ratio↑	Pene.Depth↓	Pene.Vol.↓
GT	-	-	-	1.	0.269	12.08
CosHand [69]	50.168	50.149	45.135	0.685	0.606	24.496
Afford-diff [82]	37.76	37.68	34.07	0.553	0.413	12.9
Bimart [70]	6.510	6.453	4.012	0.775	0.525	17.313
Ours	2.520	2.361	1.150	0.936	0.415	15.344

Table 2 Quantitative comparison of 3D hand pose quality on the ARCTIC dataset. Gray shading indicates the best performance. HOIDiff and HOIDiff.* are excluded here as their 3D pose accuracy is already represented by the BimArt and Ours rows, respectively.

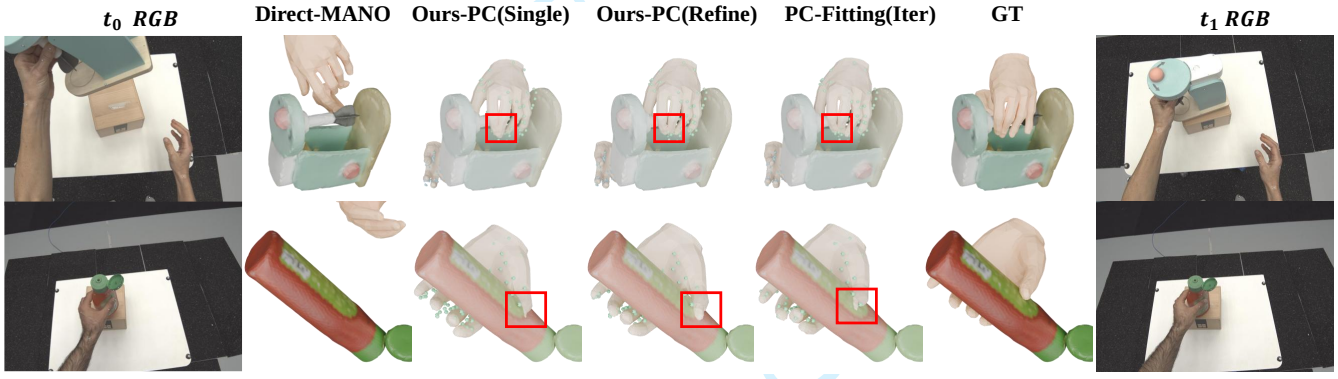


Figure 5 Qualitative comparison of different hand pose estimation and refinement strategies, PC means "Point Cloud". From left to right: (a) Input RGB at t_0, (b) Direct-MANO: direct MANO parameter prediction, (c) Ours-PC (Single): single-step analytical IK from semantic point clouds, (d) Ours-PC (Refine): refinement based on single-step initialization, (e) PC-Fitting (Iter): iterative optimization without single-step initialization, (f) Ground Truth (GT), and (g) Target RGB at t_1.

synthesis. Given a source image at the time step t and a target interaction state at the time step t', PowerPaint v2 [83] initially inpaints hand and object regions to remove occlusion artifacts and recover a static background. Subsequently, a Wan-VACE model [84] equipped with camera control mechanisms transfers this background from the source to the target viewpoint, preserving the scene layout and illumination consistency before synthesizing the target interaction foreground.

As detailed in Table 1 regarding the evaluation of 2D generation quality, the proposed approach significantly outperforms all baseline methods across all visual quality metrics. Regarding the 3D evaluation in Table 2, an inherent trade-off exists between the Contact Ratio and the penetration metrics. Nevertheless, achieving accurate contact holds a higher priority during the assessment. Physical penetration metrics lose practical meaning when the predicted hand fails to establish contact with the manipulated object.

4.3 Ablation Study

The experiments include ablation studies on the 3D geometric conditions fed into the 2D UNet branch. The objective is to evaluate which modality of 3D geometric conditions injected yields a more substantial positive impact on the 2D generation. As demonstrated in Table 3, injecting depth maps as the 3D geometric condition leads to the most significant improvement. This finding aligns with the initial hypothesis that depth maps provide explicit and rich spatial information to guide the 2D generation process effectively.

The evaluation validates the effectiveness of the single-step inverse kinematics approach utilizing semantic point

Method	PSNR↑	SSIM↑	LPIPS↓	MSE↓	MAE↓	FID↓	sFID↓	CLIP-S↑	IS↑
w/o cold-start	15.431	0.544	0.290	0.037	0.103	40.232	0.164	18.934	3.981
w/o depth	16.203	0.578	0.253	0.034	0.091	29.550	0.111	19.147	4.075
w/ normal	16.083	0.574	0.265	0.034	0.094	35.787	0.166	19.402	3.911
Ours(w/ depth)	16.187	0.580	0.252	0.033	0.091	28.183	0.103	19.186	4.137

Table 3 Ablation study on 2D generation quality. Gray shading indicates the best performance. "w/o cold-start" denotes the exclusion of initial hand priors, relying solely on object geometry. The results demonstrate that incorporating depth maps as a 3D geometric condition yields the most significant improvement in visual fidelity.

Method	MPVPE↓	MPJPE↓	CD↓	Contact Ratio↑	Pen.Depth↓	Pen.Vol.↓	pFID↓	time ↓
Only cold-start	4.777	4.774	2.113	0.915	1.061	3.467	0.055	-
Direct-MANO	12.716	12.581	8.241	0.648	0.390	0.914	0.746	-
PC-Fitting (Iter)	2.550	2.461	1.167	0.934	0.446	15.679	0.022	5.94
Ours-PC (Single)	2.640	2.520	1.216	0.918	0.432	15.574	0.022	0.017
Ours-PC (Refine)	2.520	2.361	1.150	0.936	0.415	15.344	0.022	3.39

Table 4 Ablation study of 3D hand pose quality on the ARCTIC dataset. Best results are highlighted in gray. We evaluate various 3D diffusion input conditions and hand pose estimation strategies, with naming conventions consistent with Figure 5. The results demonstrate the effectiveness of joint training with the 2D appearance branch, the advantages of using semantic point clouds as the prediction modality, and the superior efficiency of the proposed single-step analytical IK approach.

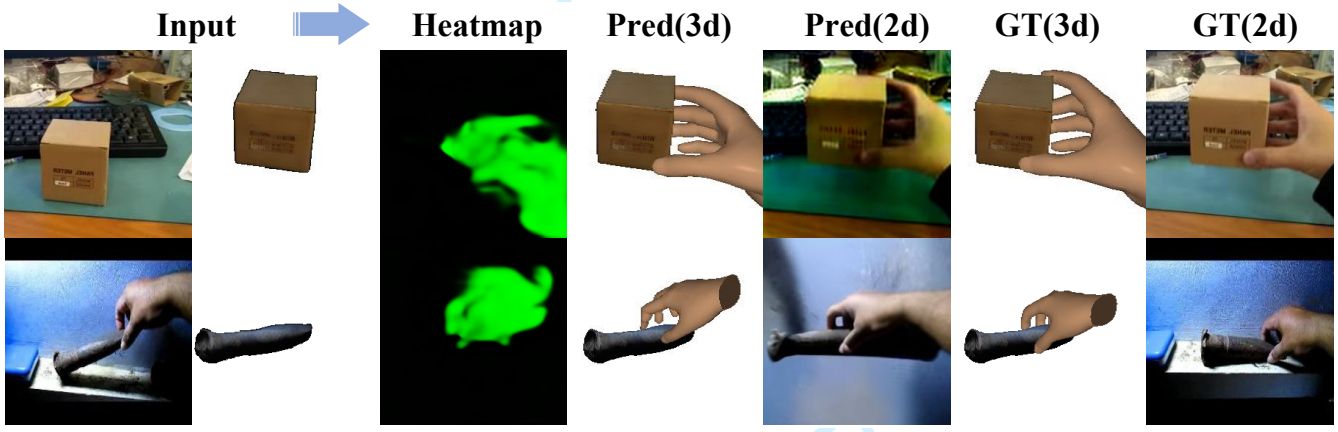


Figure 6 Qualitative results of THREAD on the WildO2 dataset. This figure illustrates the joint 2D and 3D generation outcomes for in-the-wild objects, specifically demonstrating interactions with a box and an iron rod.

clouds. As shown in Table 4, leveraging the specialized semantics of the predicted point clouds to directly infer a usable set of MANO parameters in a single step operates significantly faster than the traditional iterative fitting method. While both approaches infer parameters from predicted point clouds, the single-step method avoids computationally expensive cyclic optimization. This speedup incurs only a marginal degradation in quantitative metrics. Building upon the initialization provided by the single-step approach, the subsequent refinement stage further enhances the generation quality.

An additional experiment demonstrates that optimizing intermediate point cloud representations outperforms the direct optimization of MANO parameters, which is susceptible to local optima due to high degrees of freedom and non-linear skinning processes. Providing direct gradient information, point cloud optimization avoids the complex loss weight balancing required for MANO components, including shape, axis-angle representations, and global translation. Since these MANO parameters reside in optimization spaces distinct from other 3D scene inputs, direct optimization often causes the generated hands to float in the 3D space, as illustrated in Figure 5.

4.4 Further Analysis

4.4.1 *Generalization to In-the-Wild Scenarios*

We evaluate the proposed approach on the WildO2 dataset [75] to demonstrate the generation capabilities of the framework in complex and diverse in-the-wild scenarios. Following the standard protocol of the dataset, the system acquires in-the-wild object templates through the 3D generation method [59]. It is noteworthy that the expected

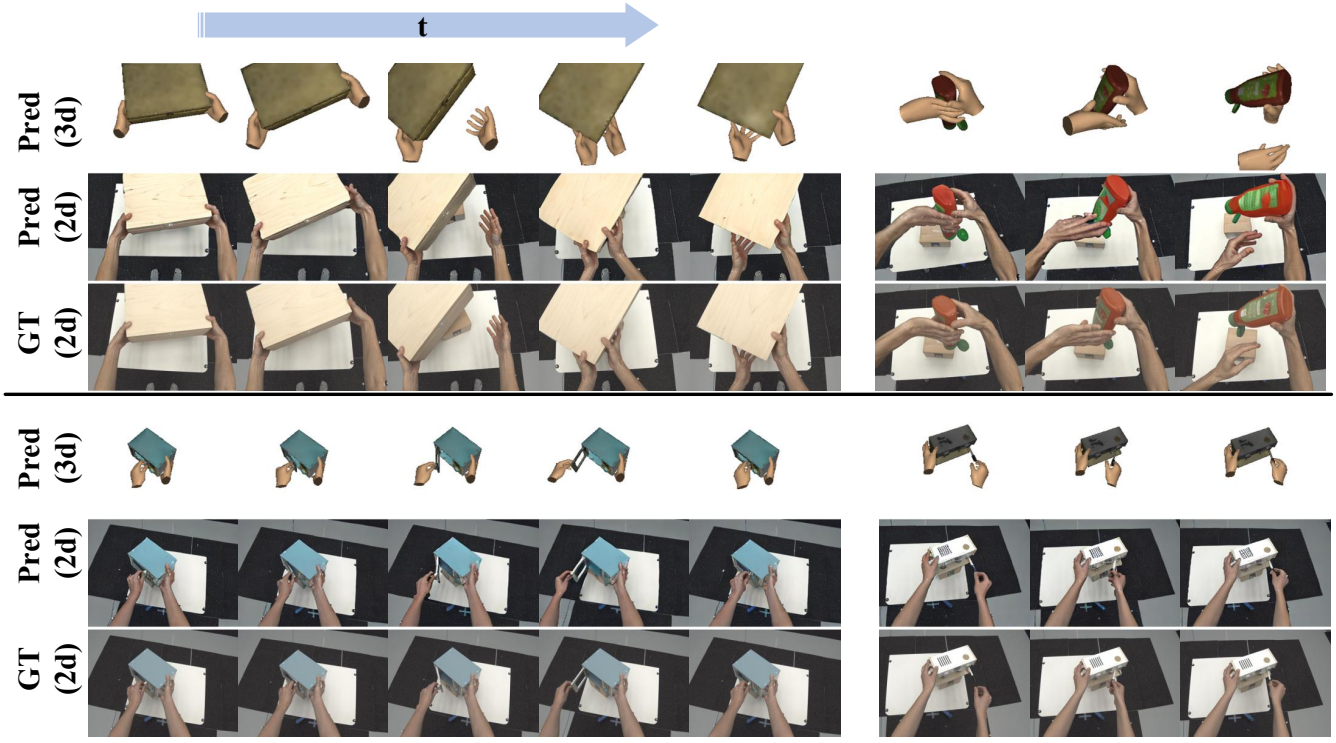


Figure 7 Qualitative results of sequence generation. THREAD synthesized relatively smooth HOI sequences by processing multiple consecutive two-frame state transitions.

Metric	box	capsulemachine	espressomachine	ketchup	laptop	microwave	mixer	notebook	phone	scissors	waffleiron
Samples	277	198	208	192	228	266	233	260	239	206	222
PSNR↑	16.439	15.670	17.457	14.268	14.308	18.965	15.591	14.108	12.376	16.152	13.680
SSIM↑	0.602	0.558	0.612	0.530	0.529	0.634	0.552	0.519	0.479	0.576	0.508
LPIPS↓	0.251	0.281	0.218	0.328	0.327	0.168	0.280	0.331	0.409	0.255	0.348
MPJPE↓	2.150	2.230	1.820	2.823	2.651	1.724	2.235	2.752	3.901	2.302	2.861
Contact Ratio↑	0.960	0.891	0.956	0.917	0.921	0.973	0.906	0.947	0.913	0.944	0.937
Pen.Depth ↓	0.253	0.419	0.414	0.7403	0.403	0.610	0.448	0.538	0.271	0.233	0.403

Table 5 Category-wise sample counts and statistics for 2D image quality and 3D interaction quality on the ARCTIC dataset.

states between the object-only frame and the exact hoi frame in this configuration involve minimal pose variations of the object. This subtle visual discrepancy significantly impacts the quality of the interaction heatmaps predicted by THREAD. Despite these challenges, the framework successfully synthesizes plausible joint 2D&3D HOI by leveraging the robust priors of large-scale 2D pre-trained models alongside the geometric guidance provided by the 3D generation module, as illustrated in Figure 6.

4.4.2 *Sequence Generation*

While preceding evaluations focus on single-state generation, our method demonstrates the potential to be extended to sequence generation for practical applications. Although our pipeline lacks explicit temporal supervision, its cross-frame coherence primarily stems from the explicit 3D modeling, which provides consistent geometric priors. To evaluate this, we sequentially process representative ARCTIC sequences via multiple two-frame state transitions, as depicted in Figure 7. Because the framework fundamentally operates on single frames, occasional discontinuities, such as abrupt hand variations or object artifacts, inevitably emerge. Nevertheless, it successfully synthesizes relatively smooth interaction sequences in most scenarios. Furthermore, our framework demonstrates strong extensibility. The single-frame 2D and 3D backbones can be readily upgraded with temporal modules to introduce explicit temporal supervision, seamlessly extending the pipeline to video synthesis and 4D motion generation tasks, respectively. [52] Since the 3D module merely provides an initial prior and does not continuously influence the 2D synthesis process, the final 3D and 2D outputs may lack strict structural alignment.

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4.4.3 *Cross-Category Analysis*

Table 5 details the performance variations across different object categories within the ARCTIC dataset. In 2D generation, performance degrades significantly for the "phone" and "waffleiron" categories, while "espressomachine" and "microwave" achieve satisfactory results. This discrepancy correlates strongly with predominant interaction patterns: the former involve large-scale dynamic movements (e.g., folding), whereas the latter primarily entail localized manipulations (e.g., toggling levers or opening glass doors) that minimally deform the object's overall geometry. For 3D generation, the "capsulemachine" and "mixer" categories yield sub-optimal Contact Ratios, as they often involve fine-grained manipulations. The poor Mean Per Joint Position Error (MPJPE) observed for "phone" and "waffleiron" in 3D generation aligns tightly with their degraded 2D quality, underscoring the substantial impact of the 2D prior on final 3D outcomes.

5 Conclusion

In this paper, we presented THREAD, a unified generative framework that couples 2D appearance synthesis with 3D motion prediction for first-person hand-object interactions. A prior-guided cold-start mechanism resolves the cyclic dependency between appearance synthesis and pose estimation, while the bidirectional 2D-3D coupling via score distillation and explicit affordance constraints promotes physical consistency and macroscopic scene understanding. The proposed semantic point representation, paired with a non-iterative inverse kinematics algorithm, further eliminates the computational burden of iterative optimization while preserving structured physical constraints. Extensive experiments demonstrate that THREAD effectively leverages large-scale 2D visual knowledge to guide complex 3D synthesis across diverse contexts.

Limitations and future work. While THREAD achieves promising results, several limitations remain for future investigation. The current framework primarily focuses on short-term interactions, and extending it to long-horizon, multi-step manipulation sequences is a significant challenge. Additionally, the scalability to highly complex or deformable geometries and the strict structural alignment between the 3D and 2D outputs warrant further exploration.

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